What a time-axis read-out buys is temporal order

A difference-in-differences on the GAIN, on SHD. Seed-paired, n = 32 at the anchor, h128 / published-2ms / d32-L4 / e400.

NOT SHOWN HERE, AND NOT CLAIMED: that SHD depends on temporal order. That is established — Cramer et al. 2022 could not exceed 60% on spike-count-only SHD; the Neuromorphic Sequential Arena removes temporal processing model-side and reports 86.48 → 68.51; Yu et al. 2025 randomise spike times at fixed counts.
Three independent destruction operators, one conclusion, all of it prior to this work. What is measured here is WHICH COMPONENT'S contribution is the order-dependent one.

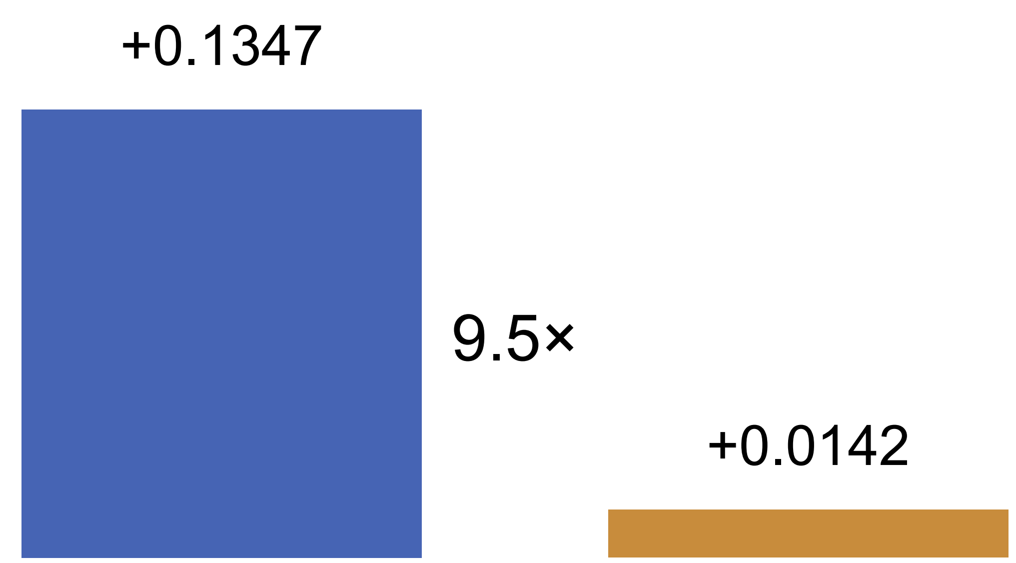
Each arm's OWN shuffle cost — the pair is the measurement

The operation — on the INPUT

permute time bins,
independently per sample,
in BOTH the training and
test splits

the task becomes
rate-solvable

Nothing is removed from the model.
This is not an ablation of the read-out,
and no arrow here points at a component.



time-axis attention
d32 / L4 read-out
32 / 32 seed pairs positive

rate read-out
ff+fixed — half the measurement,
not a faint control

Measured across the design space

21 of 21 operating points carry
the control, and EVERY ONE
clears its +0.03 bar

widths 128 → 1024,
both contracts, both geometries

all at e400 AND e100;
depth is a scope limit: h768

The read-out's advantage over the rate arm falls +0.1275 → +0.0070: 94.5% of it is contingent on temporal order.
That is the collapse of the ADVANTAGE, which is not the gain and not a share of it — see the strip below.

THE EFFECT'S SIZE IS NOT THE GAIN. Spearman ρ between the six per-width gains and their difference-in-differences is -0.1430, against a registered bar of +0.829.
h768 carries the SMALLEST positive gain on the ladder (+0.0560) and the LARGEST difference-in-differences in the campaign (+0.1881).
The claim is that the read-out's contribution IS order-dependent. It is NOT that the gain decomposes into an order-dependent share and a remainder.

Accuracy, for scale and no more: the read-out reaches 0.8332 against the rate arm's 0.7057. It is NOT COMPETITIVE — the SHD frontier is 95.07–96.3% via learned delays, adaptation and a state-space model, and this instrument carries no temporal kernel of any kind. No accuracy claim is made and none is drawn larger than the costs above.

SCOPE, on the image because an abstract travels alone: the gain INVERTS at width h1024 (-0.1618 at d32/L4), the collapse there is LOCATED BUT UNEXPLAINED, and 0.80 clearance is geometry-specific. Three preregistered rescue levers at h1024 all failed and every one is worse than the arm it was meant to rescue.

No biology, no cortex, no neuromorphic-hardware claim, and no claim that attention makes a network temporal.